

SolarMax

Flashing the ESP32 — Step-by-Step Guide

VS Code + PlatformIO. Beginner-friendly — works on **Windows** and **Mac**.

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1. Before you start

You need these on hand:

- A computer (Windows or macOS) with internet.
- The ESP32 DevKit V1 board.
- A **USB data cable** (Micro-USB) that fits the board.

Note

This guide works on **Windows** and **Mac**. The steps are the same on both; a blue **On Windows** box appears wherever something differs (the installer, the USB driver, and the port name). The screenshots are from a Mac, but VS Code looks the same on Windows.

Important

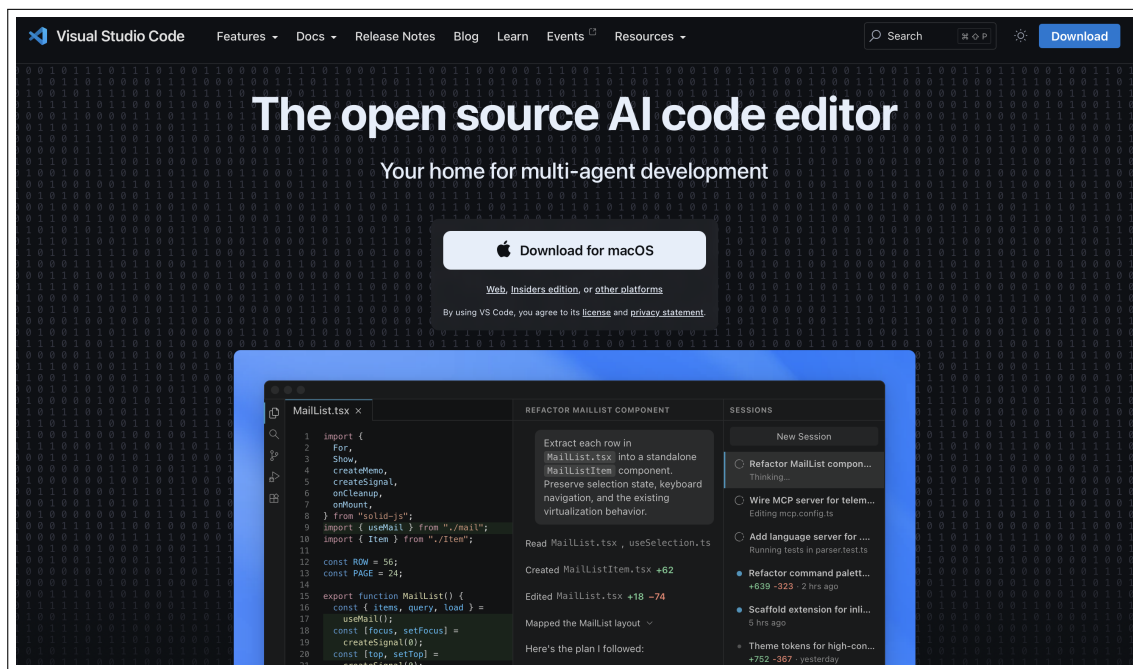
Many USB cables are **charge-only** and carry no data — the board will power on but the computer won't see it. If your board is not detected later, the cable is the #1 suspect. Use a known data cable (e.g. one that syncs a phone).

[IMAGE PLACEHOLDER: board-overview.jpg]

Screenshot to take: A clear photo of the ESP32 board, with the USB port, the EN (reset) button, and the BOOT button visible.

2. Install Visual Studio Code

1. Go to code.visualstudio.com.
2. Click the big download button for your system (Windows or Mac).
3. Open the downloaded file and follow the installer (accept the defaults).
4. Launch VS Code.



*The Visual Studio Code download page. Click the large download button for your system (or the **Download** button at the top-right).*

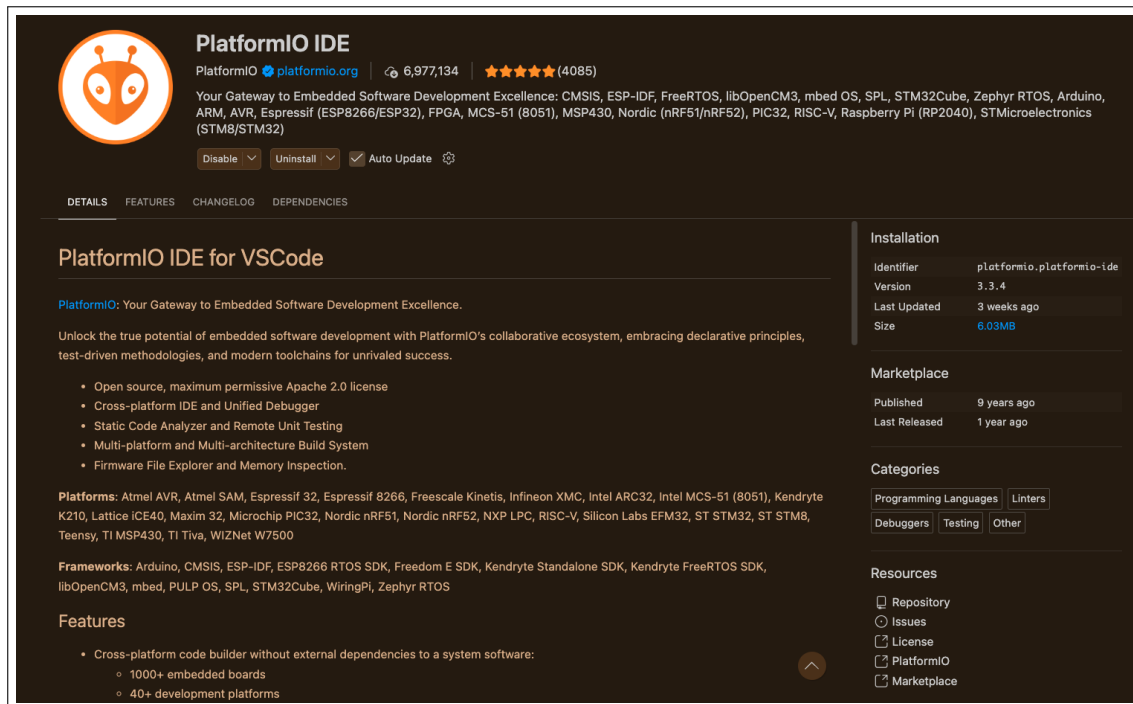
On Windows

Click **Download for Windows** — you'll get a `.exe` installer. Run it and accept the defaults (it's fine to leave "Add to PATH" checked).

3. Install the PlatformIO extension

1. In VS Code, click the **Extensions** icon in the left bar (four squares).
2. In the search box, type PlatformIO IDE.

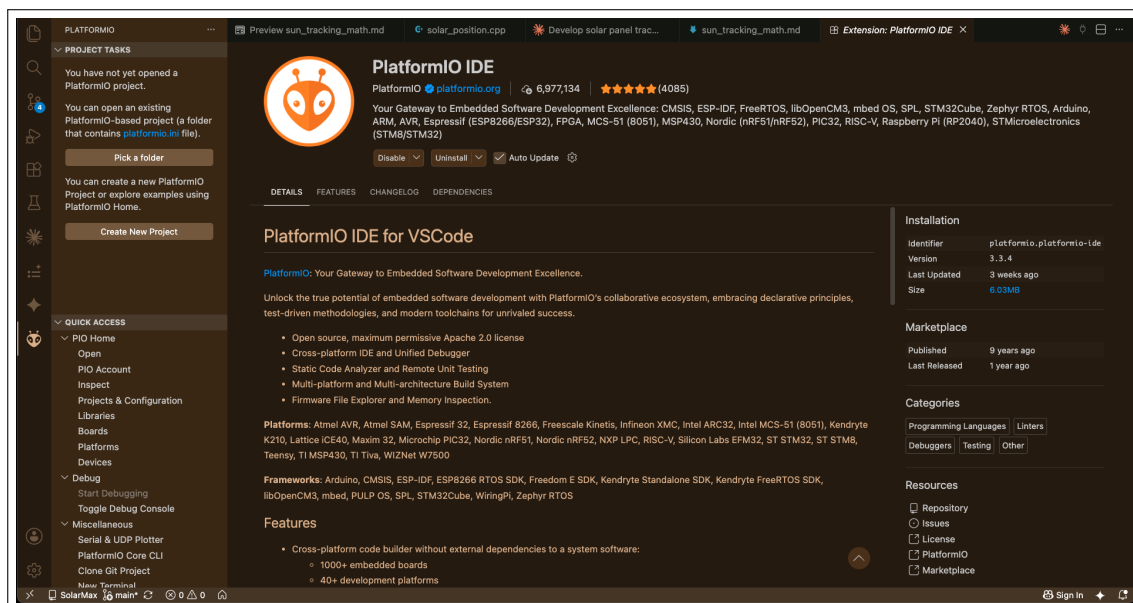
3. Click **Install** on the “PlatformIO IDE” result.
4. Wait for it to finish — this can take a few minutes. When prompted, allow it to reload/restart VS Code.



The PlatformIO IDE extension page. Click **Install**. (Here it is already installed, so it shows *Disable/Uninstall* instead.)

Tip

When PlatformIO finishes installing, a small **alien-head icon** appears in the left bar. Clicking it opens **PlatformIO Home** — that’s how you know the install worked.

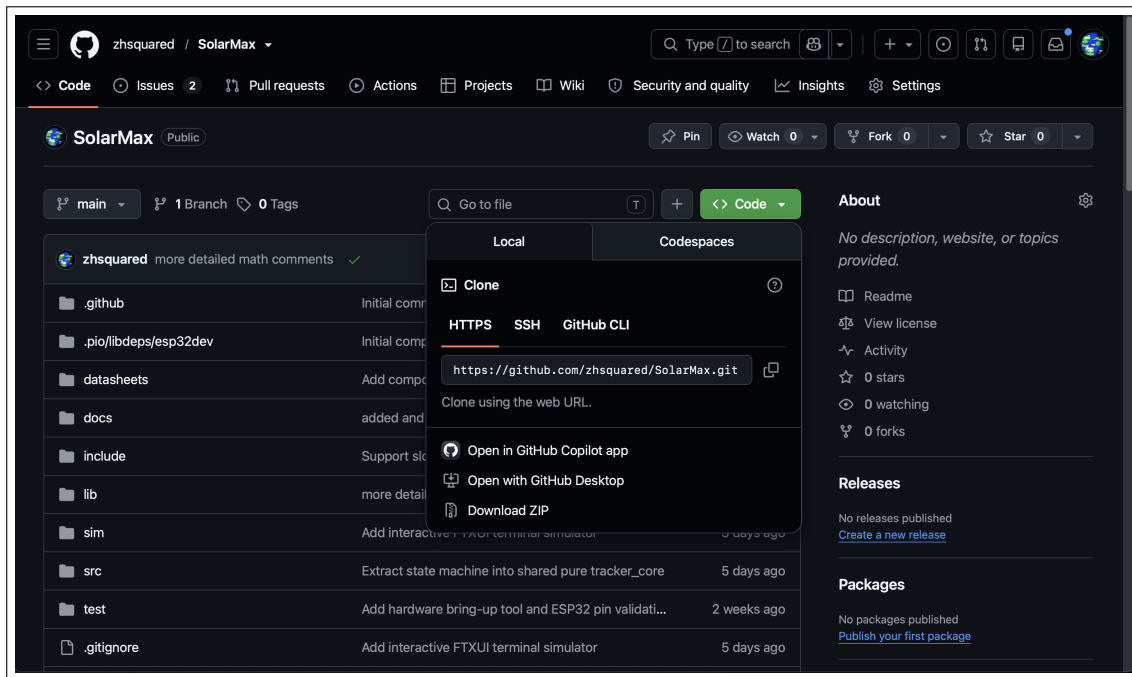


PlatformIO Home. The alien-head icon in the far-left bar (highlighted near the bottom) opens it. From here you can “Pick a folder” to open a project.

4. Get the SolarMax project onto the computer

Pick *one* option:

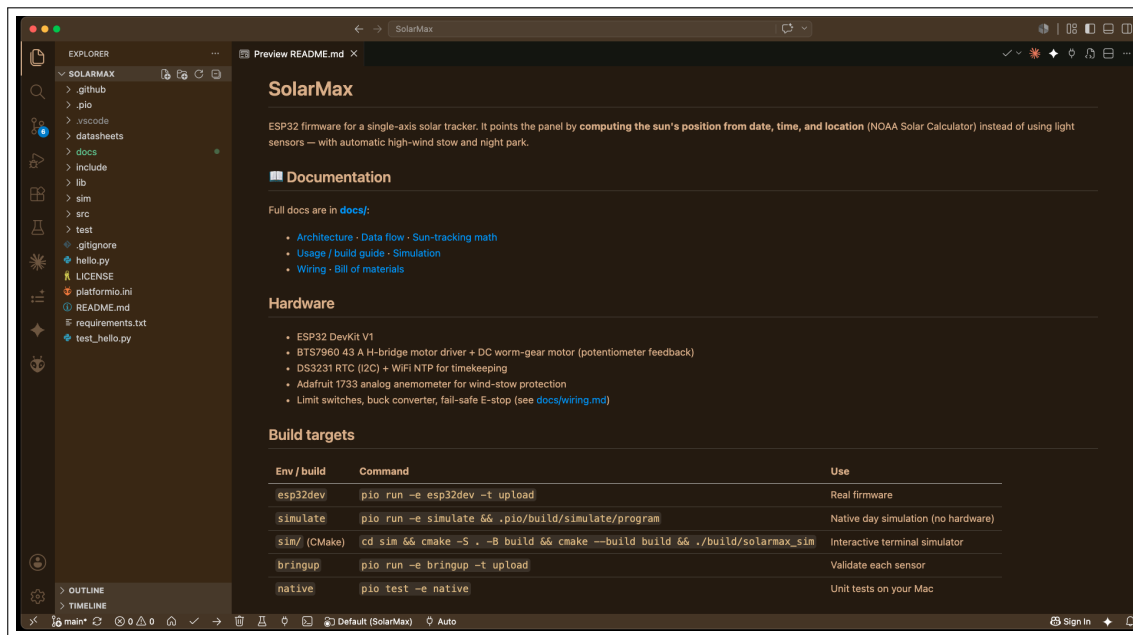
- **Easiest — download a copy:** open github.com/zhsquared/SolarMax in a browser, click the green **Code** button, choose **Download ZIP**, then unzip it. You’ll get a folder named **SolarMax**.
- **Or copy it:** get the **SolarMax** folder from a teammate (USB drive or shared drive).



On the GitHub repo page, click the green **Code** button and choose **Download ZIP** (or copy the HTTPS clone URL shown).

5. Open the project in VS Code

1. In VS Code: **File** → **Open Folder...**
2. Select the **SolarMax** folder (the one that contains **platformio.ini**) and open it.
3. Wait. The first time, PlatformIO downloads the ESP32 tools automatically — a progress bar shows at the bottom. Let it finish before continuing.

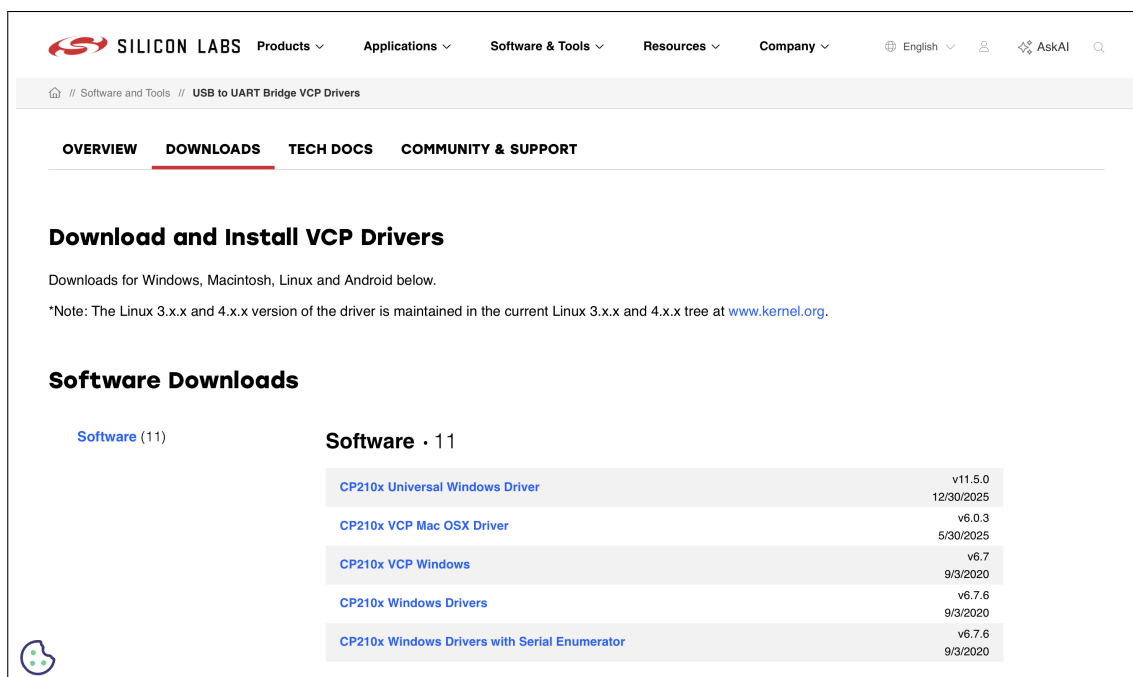


The SolarMax project open in VS Code. The Explorer on the left lists *platformio.ini* plus the *src*, *lib*, *include*, *docs*, and *sim* folders.

6. Install the USB driver (only if the board isn't detected)

Most computers already have this. If, during upload, no port is found, install the driver for your board's USB chip:

- **CP210x** (most DevKit V1 boards): [Silicon Labs CP210x VCP driver](#).
- **CH340** (some clones): search “CH340 driver” for your OS.



The Silicon Labs CP210x driver downloads. Choose the one for your operating system (e.g. “CP210x VCP Mac OSX Driver”, or a Windows driver).

On Windows

Windows often needs this driver. Download the **CP210x Universal Windows Driver**, **unzip** it, then right-click the file `silabser.inf` and choose **Install** (or run the included `CP210xVCPInstaller_x64.exe`). Then open **Device Manager**: under **Ports (COM & LPT)** you should see “Silicon Labs CP210x...(COM x)”.

7. Plug in the ESP32

Connect the board to the computer with the USB *data* cable. A power LED on the board should light up.

[IMAGE PLACEHOLDER: board-plugged-in.jpg]

Screenshot to take: The ESP32 plugged into the computer, with its power LED lit.

On Windows

To confirm Windows sees the board, open **Device Manager** → **Ports (COM & LPT)** — you should see a **COM x** entry (e.g. COM3). No entry means the driver isn’t installed yet (Section 6).

8. Upload (flash) the firmware

1. Look at the **toolbar along the very bottom** of the VS Code window — a thin dark bar with small icons (this is the PlatformIO toolbar).
2. Click the **right-arrow icon** (→) — “PlatformIO: Upload”. (The checkmark next to it is Build-only; the arrow both builds *and* uploads.)
3. The terminal at the bottom compiles the code and flashes the board. Wait for it.
4. Success looks like [SUCCESS] and “Took ...seconds” in green.



*The PlatformIO toolbar at the bottom of the window. Left to right: home, checkmark (**Build**), right-arrow (**Upload**), and further right the plug icon (**Serial Monitor**). “Default (SolarMax)” shows the active project.*

[IMAGE PLACEHOLDER: vscode-07-upload-success.png]

Screenshot to take: The VS Code terminal showing a successful upload ending in [SUCCESS].

Important

If upload fails with a “Connecting...” timeout: hold the board’s **BOOT** button while it says “Connecting”, then release it once it starts writing. Also double-check the cable (Section 1) and the driver (Section 6).

9. Watch it run (Serial Monitor)

1. In the same bottom toolbar, click the **plug icon** (“Serial Monitor”).
2. You should see the SolarMax start-up banner and the self-check report.

[IMAGE PLACEHOLDER: vscode-08-serial-monitor.png]

Screenshot to take: The VS Code Serial Monitor showing SolarMax boot output (the “SolarMax” banner and the self-check lines).

Tip

That’s it — the ESP32 is flashed. To flash again after code changes, just click the Upload arrow again.

10. Troubleshooting

- **No port / board not detected:** try a different (data) USB cable; install the USB driver (Section 6); try a different USB port.
- **Stuck on “Connecting...”:** hold **BOOT** during the connect step, release when writing starts.
- **Nothing on Serial Monitor:** make sure the baud rate is **115200** and the correct port is selected.
- **Upload works but behaves oddly:** press the board’s **EN** (reset) button once to restart it.
- **(Windows) Which port is it?** Open **Device Manager** → **Ports (COM & LPT)** to see the board’s COM number. If nothing appears there, the USB driver isn’t installed (Section 6).